Three-chromatic Cayley graphs of exponent-four groups

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Abstract

We answer Question 5.3 of Krebs and Sankar. A nonabelian group of exponent four admits an undirected Cayley graph of chromatic number three if and only if it is not a direct product of the quaternion group of order eight and an elementary abelian two-group. The classification includes infinite groups, and every positive case has a connection set of at most ten elements. We give a direct structural proof and a complete Lean formalization of the classification.

1 Introduction and the dihedral witness

Krebs and Sankar classified the abelian groups admitting a three-chromatic Cayley graph and asked for the corresponding classification among nonabelian groups of exponent four [1, Question 5.3]. The obstruction turns out to be precisely the quaternionic family. An explicit eight-vertex graph supplies all positive cases through sections, while a graph isomorphism reduces the negative cases to Payan's theorem on cubelike graphs [2].

For a group G and $S = S^{-1} \subseteq G \setminus \{1\}$, we join x to y in $\mathrm{Cay}(G, S)$ when $x^{-1}y \in S$. In particular, the graph need not be connected, as in the question under consideration. We work in ZFC. A *section* of G means a quotient H/N of a subgroup $H \leq G$ by a normal subgroup $N \trianglelefteq H$. We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

Theorem 1. *Let G be a nonabelian group in which $g^4 = 1$ for all $g \in G$. The following are equivalent:*

1. *G admits a simple undirected Cayley graph of chromatic number three;*
2. *G has a section isomorphic to the dihedral group D_8 of order eight;*
3. *G is not isomorphic to $Q_8 \times E$ for any elementary abelian two-group E.*

The group E may be infinite. In (1), the connection set can be chosen to have at most ten elements.

The section criterion also appears in the public Lean development [6], conditional on three mathematical inputs, including a finite structural classification. That development verifies the same dihedral witness used below and the transfer through sections. Our proof establishes the structural implication directly for arbitrary groups, without a finite classification or a local-finiteness theorem, and gives a connection set of uniformly bounded size. Goldstone and Weld [3] characterized the same quaternionic family by a different property, namely that inversion preserves every undirected Cayley graph. The final direct-product splitting also follows from McCabe and Weld [4, Proposition 4.10]

once centrality of involutions and a common square for elements of order four have been established. We include that step to keep the argument self-contained.

Put $D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$ and $S_0 = \{r^2, s, rs\}$. These are three involutions. With vertices ordered as $1, r, r^2, r^3, s, rs, r^2s, r^3s$, respectively, the assignment

$$(0, 0, 1, 2, 1, 1, 2, 0)$$

is a proper three-coloring of $\mathrm{Cay}(D_8, S_0)$. The vertices $1, r^2, r^2s, r, rs, 1$ form a five-cycle. Hence the graph has chromatic number three; it is the eight-vertex Wagner graph. This concrete connection set and its three-chromaticity are also verified in [6].

Lemma 1. *If G has a D_8 section, it admits a three-chromatic Cayley graph with at most ten elements in its connection set.*

Proof. Let $\pi : H \twoheadrightarrow D_8$ be the quotient map and let $q_0, q_1, q_2, q_3, q_4, q_5 = q_0$ be the displayed five-cycle. Choose $v_i \in \pi^{-1}(q_i)$ for $0 \le i < 5$, set $v_5 = v_0$, and put

$$S = \{v_i^{-1} v_{i+1},\ (v_i^{-1} v_{i+1})^{-1} : 0 \le i < 5\}.$$

Its image under π lies in S_0, so $S = S^{-1}$ and $1 \notin S$. The three-coloring of the dihedral graph pulls back to $\mathrm{Cay}(H, S)$, while $v_0, v_1, v_2, v_3, v_4, v_0$ is an odd cycle. Regard S as a subset of G. Every component of $\mathrm{Cay}(G, S)$ lies in a left coset of H; translate the coloring across the cosets. Thus $\chi(\mathrm{Cay}(G, S)) = 3$. □

2 A structural dichotomy

We give a direct argument without a finiteness assumption.

Lemma 2. *If a nonabelian group G of exponent four has no D_8 section, then $G \cong Q_8 \times E$ for an elementary abelian two-group E.*

Proof. Two noncommuting involutions generate D_8: their product has order four and they act as two reflections. Hence all involutions of G commute. The elements of square one form a normal elementary abelian subgroup T.

We first show that every involution is central. Otherwise, choose $t \in T$ and $x \in G$ such that $t' = xtx^{-1} \neq t$. Necessarily x has order four. Conjugation by x has order at most two on T, since $x^2 \in T$ and T is abelian; it interchanges t and t'. Thus $A = \langle t, t' \rangle \cong C_2^2$ is normal in $H = \langle A, x \rangle$. Write $w = x^2$. If $w \in A$, it is the unique nonidentity element of A fixed by the swap, namely tt'. Then H has order eight and is D_8. If $w \notin A$, the group H has order sixteen and $H/\langle w \rangle \cong A \rtimes C_2 \cong D_8$. Both alternatives contradict the hypothesis.

All squares are therefore central. Quotienting by the central subgroup generated by squares produces a group of exponent two, hence an abelian group. Thus G has nilpotence class at most two. For all $x, y \in G$, the commutator $[x, y]$ is central and $[x, y]^2 = [x^2, y] = 1$.

Take noncommuting x, y and write $a = x^2$, $b = y^2$, $c = [x, y]$, $C = \langle a, b, c \rangle$, $H = \langle x, y \rangle$. Here C is a central elementary abelian two-group, $c \neq 1$, and $H/C \cong C_2^2$: the commuting involutions xC and yC generate the quotient, and they are nontrivial and distinct, since otherwise x and y would commute. If $a \neq c$, there is a linear functional $\ell : C \to C_2$ with $\ell(c) = 1$ and $\ell(a) = 0$; the same holds with b in place of a if $b \neq c$. The quotient $H/\ker \ell$ is well-defined because $\ker \ell \le C \le Z(H)$. It has order eight, nontrivial commutator, and a noncentral involution (the image of x or y). It is D_8. Consequently every noncommuting pair satisfies

$$x^2 = y^2 = [x, y] \neq 1. \tag{1}$$

Choose noncommuting i, j and set $z = i^2 = j^2 = [i, j]$. Equation (1) gives $K = \langle i, j\rangle \cong Q_8$. For every $g \in G$, each of $[i, g]$ and $[j, g]$ is either 1 or z. Conjugation by g therefore acts on K by independent sign changes of i and j; all four such automorphisms are inner in K. Hence $G = KC_G(K)$. If $h \in C_G(K)$, the pair ih, j does not commute. Applying (1) yields $(ih)^2 = z = i^2$, so $h^2 = 1$. Thus $C_G(K)$ is elementary abelian. Since it commutes with K and $G = KC_G(K)$, it is central in G. Moreover, $K \cap C_G(K) = \langle z\rangle$. Choose a vector-space complement E to $\langle z\rangle$ in $C_G(K)$. It follows that $G = K \times E \cong Q_8 \times E$. □

3 The quaternionic obstruction

The following graph isomorphism explains the exceptional family.

Lemma 3. *Every undirected Cayley graph on $Q_8 \times E$ is isomorphic to a Cayley graph on the elementary abelian two-group $\mathbb{F}_2^3 \times E$.*

Proof. Write $z = -1 \in Q_8$, and choose representatives $t(00) = 1$, $t(10) = i$, $t(01) = j$, $t(11) = k$ of $Q_8/\langle z\rangle \cong \mathbb{F}_2^2$. The bijection on vertices is

$$\phi : (z^\epsilon t(u), e) \longmapsto (u, \epsilon, e).$$

Let $S = S^{-1}$ be a connection set and put $T = \phi(S)$. For a noncentral quaternion q, we have $q^{-1} = zq$; hence $(q, e) \in S$ if and only if $(zq, e) \in S$. If two vertices have different u coordinates, the quaternion part of their difference has quotient coordinate $u + v$, while its sign may be modified by the multiplication cocycle. Membership in S ignores that sign. If they have the same u coordinate, the quaternion difference is exactly $z^{\epsilon+\eta}$. In the E coordinate, the difference is the sum because E has exponent two. Thus two vertices are adjacent precisely when the sum of their images under ϕ lies in T, proving the claim. □

Payan's theorem says that a finite cubelike graph cannot have chromatic number three [2]; see also [1, Theorem 1.1]. For arbitrary E, suppose a graph in Lemma 3 is three-chromatic. It has a proper three-coloring and an odd cycle. The finitely many E coordinates on the cycle span a finite subgroup $E_0 \leq E$. Restricting both the coloring and the connection set to $Q_8 \times E_0$ produces a finite, nonbipartite, three-colorable cubelike graph, contradicting Payan. Combining this with Lemmas 1 and 2 proves Theorem 1.

4 Formal verification

Theorem 1 has a complete formalization in Lean 4.29.0 with Mathlib, including the infinite case and the ten-element bound. The checked result is

`KrebsSankarQ53.q53_full_classification`.

Its hypotheses agree with Theorem 1. The dependency audit reports only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`.[1] The formal proof establishes the cubelike obstruction directly, whereas the proof above invokes Payan's theorem.

[1]The source, pinned dependencies, audit, and verification record are available at https://github.com/alexgocardinal/krebs-sankar-q53. Proof revision `0866eb8` passed the build and axiom audit. The combinatorial winding lemmas used in the formal cubelike obstruction are adapted from Galán's MIT-licensed development [5]; the source retains the full license notice.

Independent Python checks verify the dihedral witness and the quaternion-to-cubelike coordinate correspondence for $Q_8 \times C_2^r$, $0 \le r \le 3$. Exhaustive searches over all inverse-closed connection sets on Q_8 and $Q_8 \times C_2$ find no three-chromatic examples. These checks supplement the proofs.

References

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[6] *CayleyExpanded: Lean 4 formalization of a section criterion for three-chromatic Cayley graphs of exponent-four groups*, public repository (2026), revision `15bc128`. `https://github.com/tysunseven/CayleyExpanded`.